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Astrophysical Laboratory Exercise

Observation with 50cm Telescope and Outdoor lab cameras

Antonia Magenheim[1], Jakob Rausch[2], Christoph Greschitz[3], Andrea Tanzer[4], Alexander Scheicher[5], and Jacob Schimpl[6]

[1] e-mail: antonia.magenheim@edu.uni-graz.at, (Mat.Nr.: 11711025)
[2] e-mail: jakob.rausch@edu.uni-graz.at, (Mat.Nr.: 11702458)
[3] e-mail: christoph.greschitz@edu.uni-graz.at, (Mat.Nr.: 12011972)
[4] e-mail: andrea.tanzer@edu.uni-graz.at, (Mat.Nr.: 00325618)
[5] e-mail: alexander.scheicher@edu.uni-graz.at, (Mat.Nr.: 51908871)
[6] e-mail: jacob.schimpl@edu.uni-graz.at, (Mat.Nr.: 12021525)

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ABSTRACT

Aims. Aim of this observational lab is to get familiar with the handling and setup of the telescopes and the corresponding equipment provided by the Lustbühel observatory in Graz. In addition, it was the goal to collect experience in preparing and conducting an observation night, as well as collecting, processing and analysing observational data.
Methods. Observations were taken with the "Celestron Nexstar evo 8", the "GSO Dobson", a "Unistellar Odyssey" and a 0.5m Cassegrain mirror telescope. During the first night we tested various eyepieces to explore the changing effects on the true Field of View. On the second night we operated the 0.5m telescope and observed the exoplanet transit of HAT-P 32. For the data reduction and photometry we have additionally used the observation on the variable star RR Lyrae from the other group.
Results. The lab exercises contributed to gain insight in amateur and professional astronomy, imaging and photometry. Weather and Moon phase conditions were limiting the quality and quantity of the images from the first night. The observational data from the second night turned out to be of very low signal. Therefore the gain from this night was primarily to get a detailed instruction on how to setup and operate the 50cm telescope. The Data reduction, identification via plate solving and photometry was applied on HAT-P 32 and RR Lyrae. The light curves exhibit small variations consistent with expectations.

Key words. Observatory Lustbühel, Graz – Celestron – Dobson – 50cm telescope – RR Lyrae star – HAT-P 32

1. Introduction

For this laboratory exercise, our group conducted two observation nights at the *Observatory Lustbühel* in Graz during the winter semester 25/26. The first observation night on the 4th of November 2025 and the second night on the 14th of January 2026, both with the primary purpose of getting familiar with the handling of the instruments and the procedure before, during and after an observing night. The following report describes and summarizes the tasks which are necessary to not only prepare, setup and conduct an observing night with telescopes, but also to reduce, analyse and classify the obtained data.

The observatory provided us with three different telescopes for the first observation night and with the 50 cm telescope for the the second night. They are listed in Table 2 together with the individual telescope characteristics. In section 2.2 and section 2.3 we describe in more detail the use of these instruments and the needs of adjustments, such as exposure time, eyepiece and filter, depending on the target.

In the observational part we have learned to not only handle the equipment and choose the right combination of telescope and eyepiece, but also about alignment, target finding with the guider telescope and by hand, as well as measuring the earth rotation, seeing and limiting magnitude. The derived earth rotation, seeing and limiting magnitude are summarized in section 2.2 together with the images of the successfully resolved targets from the first observation night.

In the post-processing we have applied the common data reduction using bias, dark and flat fields (see section 3.1) and plate solving (section 3.2) to calibrate our photometry (section 3.3).

We finalise our report with a review of this laboratory exercise in section 4.

2. Observation

2.1. Preparation

Prior to the observation, we were asked to choose visible and bright objects and the information relevant for the upcoming event, such as magnitude, coordinates, best observing time, elevation above the horizon of the particular object and the present moon phase. *Simbad* was used for coordinates, brightness and type of the deep-sky objects. *Sky Safari 7 pro* was used for Moon and Saturn data. *Stellarium*

Telescope	Eyepiece	Magnification	FOV (°)
Celestron	5.5mm	370	0.22
	11mm	185	0.44
	25mm	81	0.62
	32mm	64	0.78
Dobson	5.5mm	273	0.3
	11mm	136	0.6
	25mm	60	0.83
	32mm	47	1.1

Table 1: Values for the Magnification and true Field of view in dependence of the combination of telescope and eyepiece. The Magnification and true FOV are depending on the telescopes Focal length (Celestron: 2032mm, Dobson: 1500mm), the Focal length of the eyepiece (as listed in table) and the apparent Field of view of the eyepiece (82° for the 5.5 and 11mm, 50° for 25 and 32mm), as described in (SkyWatcher 2017).

Instrument Name	Aperture (mm)	Focal Length (mm)	f-Ratio	Limiting Magnitude (mag)
Unistellar Odyssey Pro N	85	320	f/4	14.6
Celestron Nexstar evo 8	203	2032	f/10	16.5
GSO Dobson	300	1500	f/5	17.4
0.5m Cassegrain	500	4500	f/9	19.5

Table 2: A list of the telescopes and corresponding characteristics, used in this lab-exercise at the Observatory Lustbühel in Graz.

Name	RA(J2000)	Dec(J2000)	V (mag)	Type	Max Height	Best Obstime
Moon	01 52	+14 27	-12.1	M	+58 38' 14.7"	23:04
Saturn	23 46 32	-04 08 00	0.9	P	+38 57' 56.1"	20:48
Polaris	02 31 49	+89 15 51	2.0	*	+47 42' 34.1"	00:05
Altair	19 50 47	+08 52 06	0.8	*	+48 03' 36.6"	18:15
Alpheratz - Sirrah	00 08 23	+29 05 26	2.1	*	+72 10' 29.8"	21:20
Albireo	19 30 43	+27 57 35	3.1	**	+62 22' 41.0"	18:15
M45 - Plejades	03 46 38	+24 10 41	1.6	OpC	+67 08' 19.3"	00:48
M31 - Andromeda	00 42 44	+41 16 08	3.4	G	+84 20' 54.6"	21:44
M37	05 52 19	+32 32 36	6.2	OpC	+57 29' 46.8"	02:53
M57 - Ring Nebula	18 53 35	+33 01 45	8.8	PN	+60 09' 11.3"	18:15
M82 - Cigar Galaxy	09 55 52	+69 40 47	8.4	Rad	+63 30' 12.5"	05:00
NGC869 - h Persei	02 22 16	+57 08 59	6.1	OpC	+79 48' 47.3"	23:23
Aldebaran	04 35 56	+16 30 32	0.8	**	+59 30' 18.2"	01:36

Table 3: List of the targets from the Observation night from November 04/05 with coordinates, apparent visible magnitude, object type, maximum height above the horizon and the best observing time.

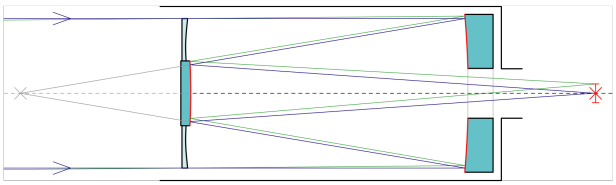
was used as a sky map and source for information about the objects. A list of the objects that we have chosen is given in Table 3. The best observing time was determined by looking for the time of the highest point above the horizon for the object during the observation night. The Observation night started at 18:20h when the sun reached 18° below the horizon and ends when it reached again −15°in height above the horizon at 05:02h. However the observing night for us ended at 22:00h, therefore we observed the Moon, Polaris, M48, M37, M82, NGC869 and Aldebaran not at their theoretically best observing time. Polaris did not change its height much during the night, therefore the quality of the observation was still guaranteed when we observed earlier than the best observation time. M57, Albireo and Altair were at higher latitude before the astronomical twilight ended. The best observation time was therefore set to the beginning of the night. For the second observation night, the star HAT-P 32, was already chosen in advance by our supervisors.

For the first observational laboratory, the observatory provided us with three different instruments. The *Celestron Nexstar evo 8* with 203 mm aperture and focal length of 2032 mm, the *GSO Dobson* with 300 mm aperture and 1500 mm focal length and the *Unistallar Odyssey Pro N 85/320 Fully Automated Smart Telescope* with 85 mm aperture und 320 mm focal length. The telescopes limiting magnitude TLM is defined as

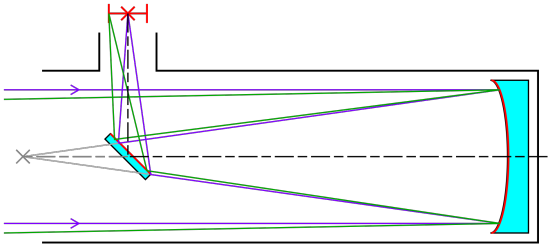
$$TLM \approx 5 + 5\log_{10}(D) + 2.5\log_{10}(F) \tag{1}$$

with the telescopes aperture D/mm and $F > 0$ combining the experience of the observer with observing conditions. The values for each telescope can be found in Table 2 assuming $F = 1$.

For the Celestron and the Dobson, we had four different eyepieces to try: the *Meade UWA 5.5mm*, the *Explore Scientific 11mm 82°*, the *25mm Super Plössl 50°* and the *32mm Plössl* eyepiece with 50° apparent field of view. The 5.5 mm and 11 mm eyepiece have both an apparent field view of 82°. The 25 mm and 32 mm have an apparent field of view with 50°. Using the telescopes with different eyepieces gives a change in the telescopic field of view (TFOV),



(a) Schmidt-Cassegrain



(b) Newtonian

Fig. 1: Comparison of the light path inside the Dobson and the Celestron Telescopes.

determined by

$$\text{TFOV} = \frac{\text{Apparent Field of View}}{\text{Magnification}} \tag{2}$$

where the apparent field of view of the eyepiece is divided by the magnification, defined by

$$\text{Magnification} = \frac{\text{Focal length of the Telescope}}{\text{Focal length of the Eyepiece}} \tag{3}$$

It is usually best to start with a lower magnification, in order to have a larger field of view. This way it is easier to find objects in the night. Once an object has been located, the eyepiece can be changed to smaller focal length, meaning a larger magnification. The values for each combination are given in Table 1.

The Dobson and the Celestron differ in their structure and optical design. While the Celestron has a Schmidt-cassegrain setup, the light in a Dobson follows a Newtonian light path. Sketches of the two setups are shown in Figure 1. The Schmidt-Cassegrain allows a longer focal length inside a small view, but the trade-off is in the light loss due to more reflections. In the Newtonian system, the light only gets reflected on two mirrors and is therefore the more 'effective' system, but comes with a comparably larger aperture.

2.2. First Observing Night: Amateur Telescopes

The first laboratory took place on the 4th of November in 2025. Our group arrived at 18:00h, which was shortly before the sun reached 18° under the horizon ("astronomical Twilight") at 18:20h (Stellarium Developers 2025). During this night the moon was in the late growing phase with 98.91% surface appearance (NASA Scientific Visualization Studio 2024). The weather conditions were mostly favourable with low cloud coverage, wind speeds close to 0km/h and temperatures of 7°C. Only the humidity of 80% was rather high and therefore affecting the seeing and limiting magnitude (meteoblue AG (2026)).

For the orientation of the individual telescopes, it was important to consider the position of the moon, since it was really bright and therefore not favourable to observe objects near the moon. In addition, objects that were further above the horizon were chosen, to avoid large atmospheric paths. The Unistellar telescope was oriented towards SW/W to capture the comet *C/2025 A6* (Lemmon) for 1 hour before it vanished behind the horizon. The Dobson was also oriented towards SW/W, away from the moon. The Celestrons orientation was adjusted throughout the whole observing night, depending on the target. But first we had to set it up correctly, as described in the following.

Usually it is necessary to collimate the secondary and the primary mirror before observation. However we had to dismiss this, since there was no working laser available. The first task was to connect the telescope with the application *SkyPortal* via Wifi. With the arrows now visible on the phone-screen, it was then possible to manually operate the telescope and adjust its orientation. Next we checked the orientation of the guider telescope based on Saturn. For this, we approximately oriented the telescope towards Saturn and then made sure it was in the middle of both, the eyepiece of the actual telescope and the small guider telescope. Then we were ready to proceed with the alignment which we started at 7:02pm. This is necessary for automatic tracking of objects in the sky as the earth rotates. Here we used the 25mm eyepiece and the Celestrons default alignment called *SkyAlign*, for which we had to choose three different bright stars in the sky and follow the steps in SkyPortal. We approximately centred the first star, *Alpheratz-Sirrah*, on the screen inside the *StarPointer* and then centred the star in the eyepiece with slower motor speed. With pressing *Align*, the first star was aligned and so we proceeded with two other stars, namely *Altair* and *Polaris*. The corresponding coordinates can be found in Table 3. For the choice of the three alignment stars it was important to consider their location on the sky to not be too close to each other, not close to the horizon and not in light polluted areas, such as near the moon for example (Celestron 2018).

Now we were ready to search for different targets, all listed in Table 3. The values for the FOV for each combination of telescope and eyepiece used is given in Table 1. Parallel to the observation with the Celestron via Skyportal, we were targeting objects with the Dobson, by hand and comparing the positions of the objects of interest with the visible star constellations on the night sky. With the Dobson telescope, the Ringnebula M57, the star cluster NGC869 and the moon were successfully resolved using the 32mm eyepiece. In order to explore the difference we have observed the same objects with the 5.5mm eyepiece. However, the Ringnebula was very faint. With the Celestron it was possible to take a picture of the target, which was captured in the lens. We have used this opportunity for the Moon, Saturn, the Ringnebula, the cigar galaxy, the Plejades and the star cluster h Persei. The images are shown in Figure 2.

Besides the setup and the observation we were asked to derive the seeing, the naked-eye limiting magnitude and the earth rotation during the night. The seeing was derived based on the appearance of the diffraction rings of Polaris through the Celestron and the 5.5mm eyepiece. According

to the pickering scale the seeing during this night was 'good' with a pickering value of 6. The FST-value (faintest star test) was derived based on the faintest star visible in the polar sequence Ursa Minor. We identified the star *Eta Ursae Minoris* (η UMi) with 4.95 mag to be the faintest. At the time of observation at 20:50h its altitude was 40°. The earth rotation was derived by measuring the drift time for the star *Aldebaran*. We have chosen to do this with the Dobson with an aperture $f_T = 300\,\mathrm{mm}$ in combination with the 11 mm eyepiece with a field stop $d_{fs} = 19.0\,\mathrm{mm}$ and apparent field of view of 82°. The true field of view, the path which the star has to travel in degrees, is given by

$$\phi = 2\arctan(\frac{d_{fs}}{2f_T}) \cdot \frac{180}{\pi} = 0.607° \tag{4}$$

which is a more precise approach than the formula in Equation 2. This value is divided by the measured drift time $dt = 150$ s, leading to an earth rotation rate $\frac{d\phi}{dt} = 0.004° s^{-1} = 349.63° d^{-1}$. In order to account for the slower earth rotation rate at higher latitudes, we divide with the cosine of the declination of *Aldebaran*: $\frac{d\phi}{dt} = 349.63° d^{-1}/\cos(16.5°) = 364.438° d^{-1}$. With this earth rotation rate the day would have a length of 24h and 19min. So in reality the earth rotates faster than what we have measured. The result can be improved with better seeing conditions, a more precise time measurement and with accounting for the fact that the star might not cross the field of view exactly through the aperture diameter.

2.3. Second Observing Night : 0.5 m Cassegrain

The second observing night started at 20:05h on the 14th of January 2026. The observations ended automatically on the 15th of January 2026 at 00:45h.

Before beginning the observations, we had to start the software using a separate PC. At first, the *OLG Observing System* has to be started. From there, we turned on the power for the individual components from right to left: *MOXA* (connection to internet), *Silex* (ethernet to USB converter), *SBIG STF-8300* (power supply for the CCD camera), *Spiegelklappen* (folds away the cover of the main mirror), *Teleskop* (power supply of the telescope itself). The dehumidifier was already turned off.

After turning on everything, we had to connect all the devices using the *SX Virtual Link* software. We had to retry that four or five times because it did not work at first. Apparently, this is normal.

The program *Automadome* is used for opening the dome and connecting it to the telescope so that it automatically turns according to the direction the telescope is pointing in.

The last part of the setup was homing the telescope i.e. telling the telescope which direction it is pointing in. For that, a program called *Autoslew* is used to start the motor and calibrate it. To calibrate the motor, we set the slewing rate to a high value, moved the telescope to south and then pressed the Home symbol.

Lastly, we started the camera software (*CCD Soft*) and used the sky map software (*TheSky6*) to go to an arbitrary bright star (as a test), took an image, moved the telescope with a small slewing rate, took another image and so on – until the star was in the centre. We also used these images to ensure that they are saved to the correct folder and to check if the telescope is focused correctly (with *SIOimage*). (Whenever our notes and memory were not sufficient, (Goldgruber & Beck 2021) was used for writing that summary.)

After that, we typed in the coordinates of the star of interest (HAT-P-32) into the sky map software, selected the V filter (green) and took an image. We tested whether the detector was saturated at the centre of the bright spot resembling the star – it was. After a few cycles of setting the exposure time and testing the saturation, we ended up with an exposure time of 30 s. We then started an image series covering the whole ingress (around 30 min), then changed the filter (and with it the exposure time) a few times as an exercise: to B (blue), to r (red), and to i (NIR including dark red). At the end, we changed the filter back to V to let it run on itself for the remainder of the transit. When the telescope arrived at the lowest possible altitude angle, it stopped the image series automatically.

The next day, Prof. Koller took bias frames, dark frames and flat fields for us and shut down the telescope.

3. Data Analysis

Data Analysis is performed in Python based on the instructions on Moodle. Essential libraries are `astropy`, `astroalign`, `astroquery` and `photutils`. The tasks are split into several chunks:

1. Data reduction
2. Plate solving and querying photometric data of field stars
3. Aperture photometry on field stars, calibrated using photometric data

3.1. Data Reduction

Data reduction is performed as detailed in the description on Moodle. First, master bias, master dark and master flat are generated via the equations below. Additionally, the master dark frame is normalized with the exposure time in order to apply it to images with different exposure times later on. Since there was only one exposure time in the dark frames, direct matching to the science images was not possible so we scaled it with exposure time instead.

$$\text{master bias} = \text{median}(\overrightarrow{\text{bias}}) \tag{5}$$

$$\text{master dark} = \text{median}(\overrightarrow{\text{dark}} - \text{master bias}) \tag{6}$$

$$\text{master dark rate} = \text{master dark}/\text{exposure(dark)} \tag{7}$$

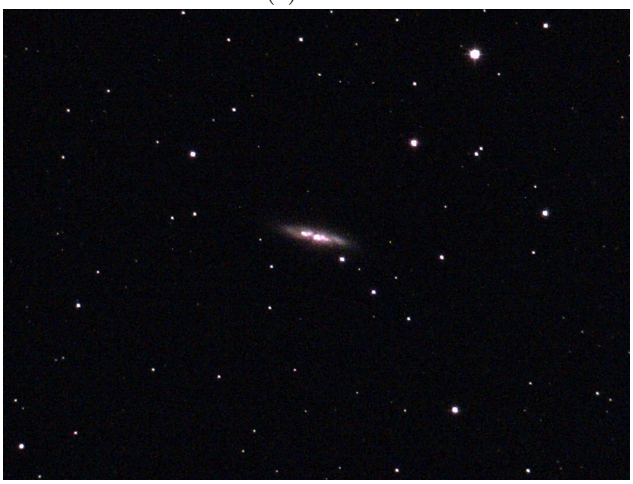
$$\text{master flat} = \text{median}(\overrightarrow{\text{flat}} - \text{master bias} - \text{master dark rate} * \text{exposure(dark)}) \tag{8}$$

Reduction of the raw frames to science frames is now straight forward:

$$\text{sci} = \frac{\text{raw} - \text{master bias} - \text{exposure(raw)} * \text{master dark rate}}{\text{master flat}} \tag{9}$$

(a) Moon

(b) M57 - Ring Nebula



(c) M82 - Cigar galaxy

(d) NGC 869 - h Persei

(e) M45 - Plejades

(f) Saturn, not very well resolved due to bad collimation.

Fig. 2

This already produces images with background noise close to zero, but since we need to subtract the median (which is close to 0) for photometry, we do it right here:

$$\text{sci} = \text{sci} - \text{median}(\text{sci}) \tag{10}$$

Figure 3 and Figure 4 show the effect of data reduction on the images of RR Lyrae. The effect is not very pronounced because of the low exposure times. It is better visible in Figure 5 at longer exposure time. Note how a few pixel are drawn to very negative values. Looking at the images, one can see that many hotpix are suppressed.

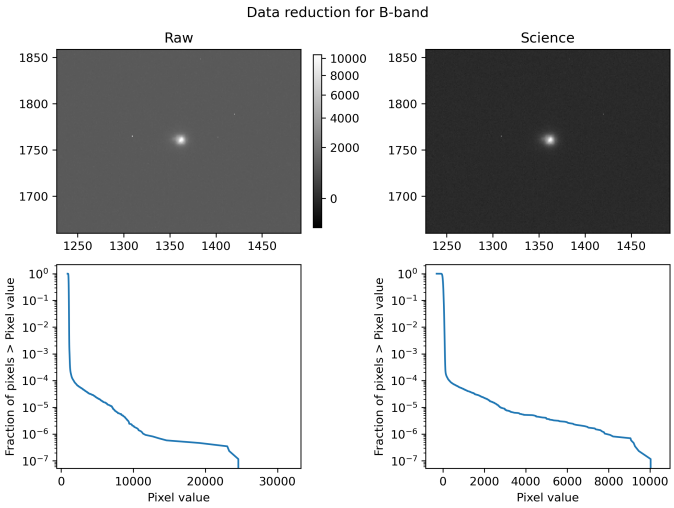


Fig. 3: Data reduction for an image of RR Lyrae in B-band. Left column: Original image. Right column: Reduced science image. Upper row: Zoom in on RR Lyrae. Lower row: Survival function showing how the background gets centred on 0 in the science frame. Additionally, all values above 10 000 are either removed or compressed into the 0-10 000 range.

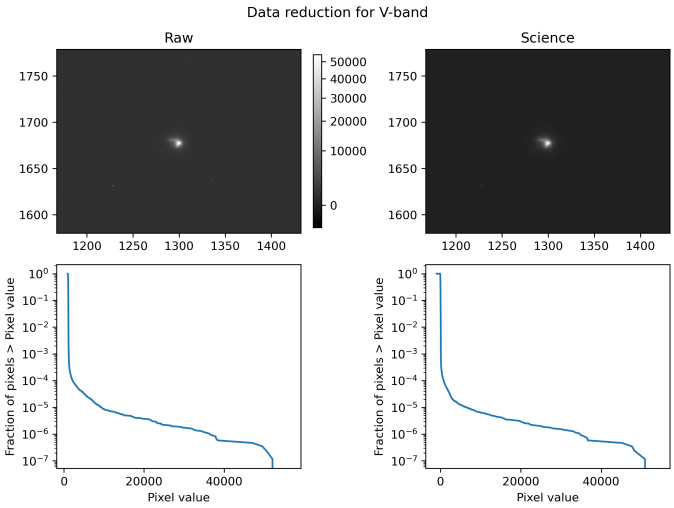


Fig. 4: Data reduction for an image of RR Lyrae in V-band. Left column: Original image. Right column: Reduced science image. Upper row: Zoom in on RR Lyrae. Lower row: Survival function showing how the background gets centred on 0 in the science frame. Here, high pixel values are barely affected. Note how distorted RR Lyrae is.

3.2. Plate Solving

Here we used the known coordinates of our targets to query for Gaia DR3 stars with $G < 13$ mag as well as UCAC4 stars with $V < 13$ mag within a radius equal to the full field of view of the telescope. This ensures that all imaged stars can be identified, even if the target is at the edge of the field of view.

The following steps have to be performed for each image individually. We apply a median blur to remove hotpix, then use `photutils.detetection.DAOStarFinder` to detect sources in our image. The parameters are fine-tuned to obtain between 11 and 30 sources since this seems to work best with `astroalign`. Unfortunately, this part is quite time consuming, leading to a total runtime of about 40 minutes if run concurrently.

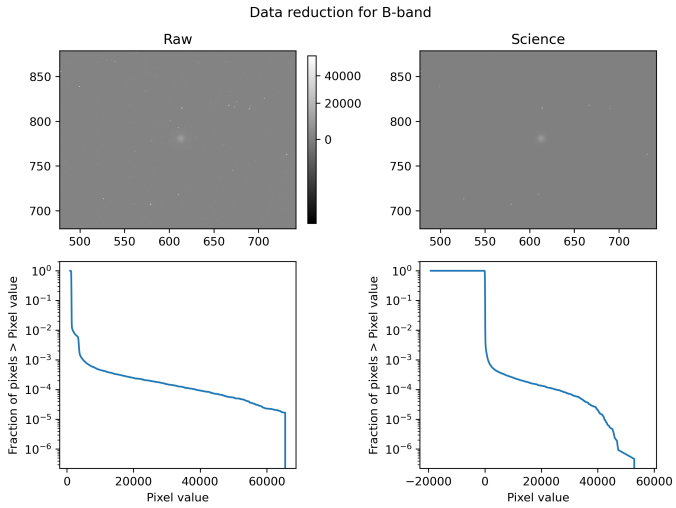


Fig. 5: Data reduction for an image of HAT-P-32 in B-band. Left column: Original image. Right column: Reduced science image showing fewer hotpix. Upper row: Zoom in on HAT-P-32. Lower row: Survival function showing how the background gets centred on 0 in the science frame. High values are clearly impacted as well.

Now we use the target coordinates to project the Gaia stars into our image. We match our sources with the Gaia stars using `astroalign.find_transform`. This allows us to obtain a WCS object using `astropy.wcs.utils.fit_wcs_from_points`, meaning the image is plate solved. In some cases this fails, so we try again using the UCAC4 table. If it still fails, we can't plate solve the image, but this mainly applies to garbage images like at the end of the RR Lyrae time series.

Finally, we obtain the coordinates of all detected sources and do a cone search with `astroquery.vizier.Vizier.query_region`. Now we know the UCAC4 identifiers of our stars, as well as their B, V, r, i magnitudes.

We create a new fits file for each image, storing the reduced image with an updated header that contains the WCS. Additionally, the star data is stored as an extension.

Since plate solving is very time consuming, it was sped up by using multiprocessing via `concurrent.futures.ProcessPoolExecutor`. This needs some additional code but reduces the runtime from 40 min per night to 10 min per night running on 7 cores.

3.3. Photometry

Before doing the photometry, we measure the size of our stars using `photutils.psf.fit_fwhm`. On 4th of November 2025 the median fwhm of all stars within each frame varied between 3 arcsec and 7 arcsec. The mean of all frames was 4.4 arcsec. During the second night on the 14th of January 2026, the fwhm varied between 2.5 arcsec and 5.5 arcsec with a mean of 3.4 arcsec. To keep our photometry self-

consistent, we settled for fwhm =3.5 arcsec for all frames. This corresponds to 7 pixels.

Next we calculate the magnitudes of our targets separately for each band. First we perform aperture photometry around each star using $r_{in} = 4$ fwhm and $r_{out} = 5$ fwhm. The mean value of the annulus is subtracted from the main aperture. The large aperture produces a light curve that is less noisy than smaller apertures. Here we normalize our fluxes with the exposure time to make the flux directly comparable between frames.

For calibration we use the known magnitudes from the UCAC4 catalogue to calculate the zero point for each star using the equation below.

$$zp = \text{true mag} + 2.5 \log(\text{flux}) \tag{11}$$

Looking at Figure 6 one can see that there is a large spread between the zero points of individual stars. There are some outliers we need to mitigate and the remaining stars still have some spread.

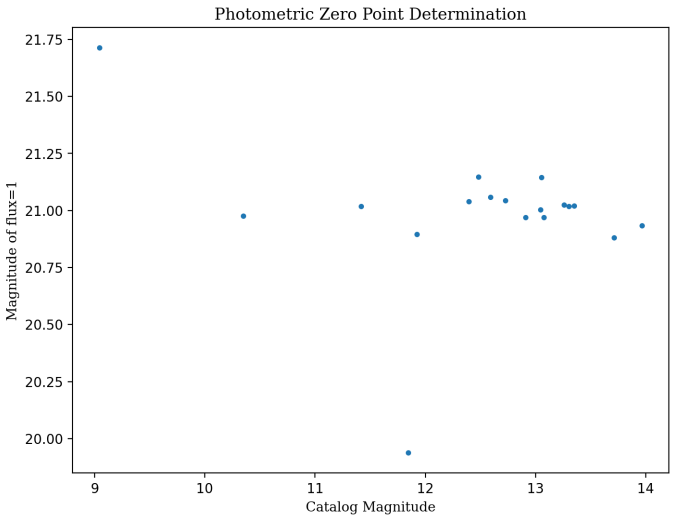


Fig. 6: Zero point determination for a single image of RR Lyrae in B-band. The star at 9th magnitude is RR Lyrae.

We tried three different ways to solve this problem. First, sigma-clipped mean with $\sigma = 1$ to remove outliers. Second, sigma-clipped mean using only those stars that appear in every image. Third, calibration using the second brightest star.

3.3.1. RR Lyrae

Figure 8 shows the light curve in V-band. Here we have the most data. Looking at the normalized flux, one can see that there is a slight decrease over time and some scatter between individual images. However, looking at the measurements, this trend is a bit weaker, especially when taking all stars into account (except outliers). Using common stars for calibration, we find a decrease of about 0.05 mag in brightness over the observation time of 2 h. With a single star for calibration, there is a similar decrease and less noise. However, this trend is less pronounced than what is indicated by the uncalibrated flux.

The reason can be found when looking at the zero points. From 18:30 to 20:30, the magnitude of the zero point increases, counteracting the decrease in flux, ie. an object with constant magnitude would produce less flux over time. A possible reason is that the airmass increased as RR Lyrae was at 70° in the beginning of the observation and at 38° in the and. Additionally, the almost full moon was rising, increasing the background brightness. Since we applied median subtraction in the data reduction, this would lead to a decrease in measured flux.

Additionally we overlay a reference light curve of RR Lyrae in Figure 8. Here, we use template 103 of (Sesar et al. 2010) for RRab stars in g and r-band to calculate the V-band template. This is scaled using a incomplete light curve from (Kloppenborg 1911) from which we extract the amplitude. Moreover, we obtain the period from (Kolenberg et al. 2011). Finally, we need the phase at the time of observation which is obtained using known peaks on October 30th and November 15th from (Borgne 2025). Looking at Figure 8, the measurements using sigma-clipped mean of all reference stars fits the actual light curve well. The other calibration methods have less scatter but are biased a bit below the expected light curve.

However, this decrease might not be significant. Figure 7 shows the light curve of a reference star that, by definition, has a constant magnitude. However, the magnitude drops by 0.2 mag over the observation time. There are other reference stars that increase in magnitude, the reason might be the intrinsic scatter of the zero point determination. RR Lyrae might be more stable because the flux is 30 times stronger..

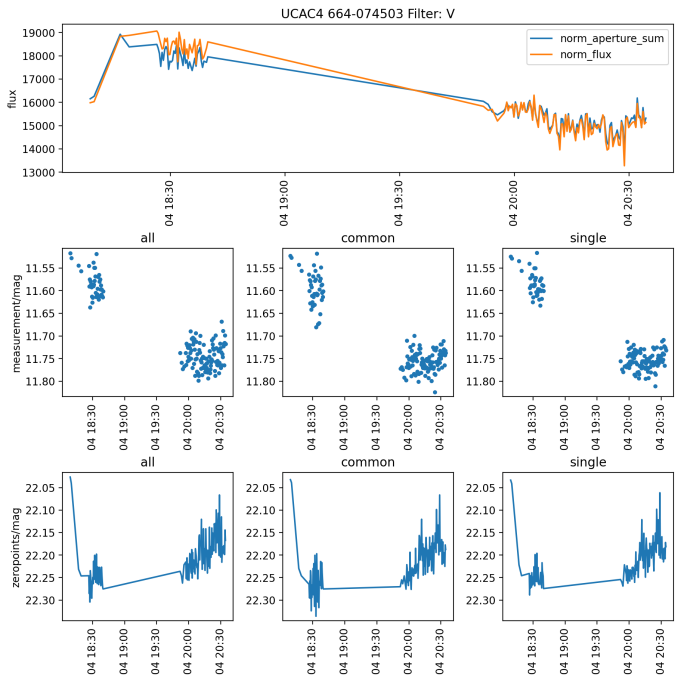


Fig. 7: Light curve of reference star UCAC4 664-074503 in V-band with V=10.171 mag. (Top): Flux normalized by exposure time without background subtraction (norm_aperture_sum) and with background subtraction (norm_flux). (Middle): Our measurements. Individual measurements are calibrated using all reference stars (all), stars appearing in all images (common) and a single star (single). (Bottom): Calculated zero points using the different methods.

Light curves using other filters are given in the appendix.

3.3.2. HAT-P-32

Analogue to RR Lyrae, we plot the light curve of HAT-P-32 in Figure 9. Towards the end of the time series there are large dips in flux, likely related to clouds. The zero points show a massive increase at these times, the light curve stays relatively constant. Here, single star calibration fails, possibly because of clouds. All stars and common stars yield almost the same result, likely because almost all stars are common stars. Unlike RR Lyrae which dominated the field, HAT-P-32 is of similar brightness as the surrounding stars, meaning they can easily be seen in all frames. The scatter is about 0.05 mag with a suggestive dip between 20:00h and 22:00h. However, the ingress started at 19:26 and was completed by 20:26h. At the end of the observation, the star was still fully eclipsed. The intensity should drop by 2.2% or 0.02 mag. Judging by the light curve, the drop would be around 5%, stronger than expected. Additionally, the light curve rises back to normal before the egress.

Light curves using other filters are given in the appendix.

4. Conclusion

This laboratory exercise gave a broad insight into the different steps and tasks necessary to absolve an Observing night with the equipment, as it is used at the Observatory Lustbühel in Graz. We completed a variety of tasks, starting at the preparation of an observing night, over the setup and alignment of telescopes, to the post-processing of the acquired observational data. The individual steps are briefly summarized in the following.

Instrumentation: We have studied the different kind of handling, setup and combinations of suitable eyepieces for a Dobson and a Celestron telescope during the first lab on November 04. However, as we anticipated a second observing night, we did not place much emphasis on ensuring that each object fully fit within the field of view. An additional laboratory exercise would help improve understanding of appropriate eyepiece selection for specific targets. We had a detailed instruction on the setup, operating and shut down for the 0.5 cm Cassegrain telescope on January 14.

Observation: We have observed numerous different objects during the outdoor lab. These objects covered the types of Planet, Stars, double Stars, Starclusters, the Moon and Galaxies. Due to the limiting weather condition we could only conduct one outdoor observing night. During this first lab the collimation of the Celestron telescope was not possible, causing a downsize to the quality of the captured images. During the second lab we have observed an exoplanet transit of the star HAT-P 32. the Signal of this star was however very weak and made it difficult to perform photometry on.

Preparation: For the target choice we have derived the best observing time for each object by its maximum height above the horizon. Due to our limited actual time at the observatory we could not observe all objects at their best observing time. Nevertheless the objects were successfully resolved. We gained initial experience in deriving important telescope characteristics such as the limiting magnitude, Magnification and true Field of View.

Theoretical Exercises: We have used the pickering scale to estimate the seeing during the first observation with pickering 6. We have derived the earth rotation rate from the drift time of Aldebaran to $364.438°d^{-1}$. Additionally the naked-eye limiting magnitude was derived using the polar sequence to a magnitude of $4.95mag$ (Eta Ursae Minoris).

Data Reduction: The data of HAT-P 32 and RR Lyrae was reduced using the Dark, Flat and Bias frames.

Plate Solving: The images where plate solved by matching GAIA star positions to sources detected in the individual images.

Photometry: We measured the flux of each star and calibrated each image using reference stars to obtain the light curves of our targets. The light curves show small variations matching our expectations. For RR Lyrae, our data fits the modelled brightness variations at that time in the V band. For HAT-P-32, they are likely insignificant.

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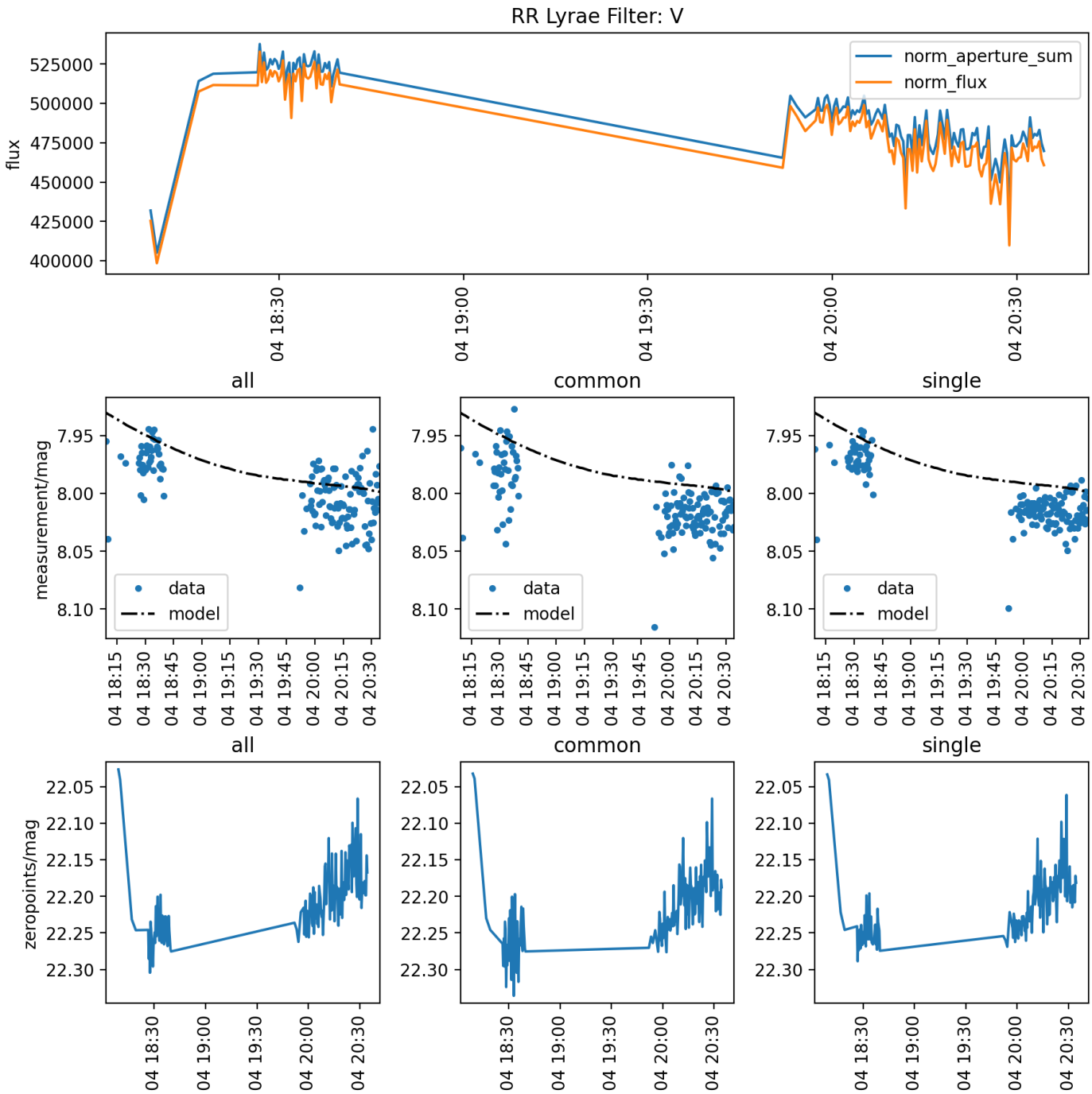

Fig. 8: Light curve of RR-Lyrae in V-band. (Top): Flux normalized by exposure time without background subtraction (norm_aperture_sum) and with background subtraction (norm_flux). (Middle): Our measurements (data) with a reference light curve (model) overlayed. Individual measurements are calibrated using all reference stars (all), stars appearing in all images (common) and a single star (single). (Bottom): Calculated zero points using the different methods.

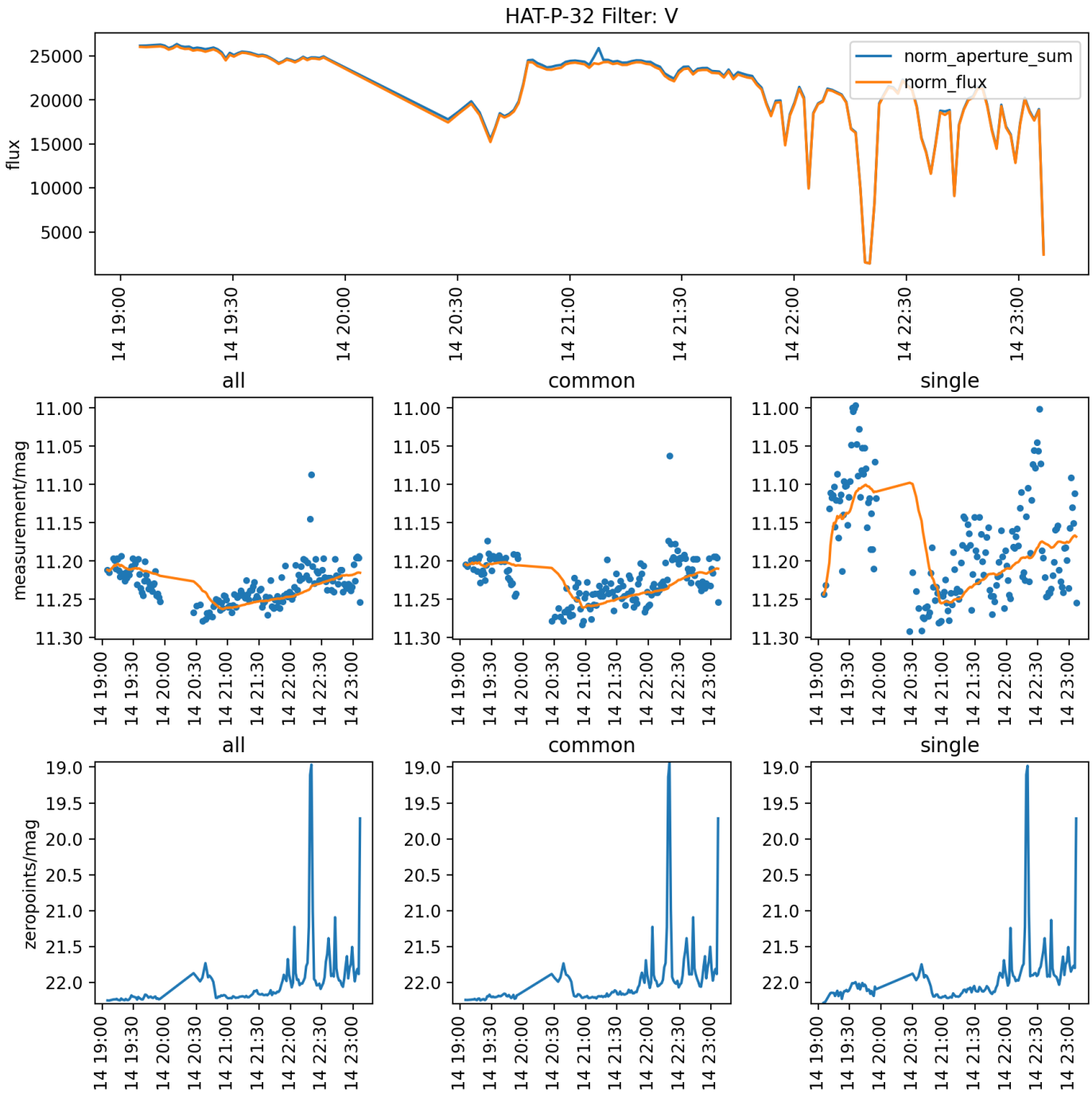

Fig. 9: Light curve of HAT-P-32 in V-band. Description see Figure 8.

5. Appendix

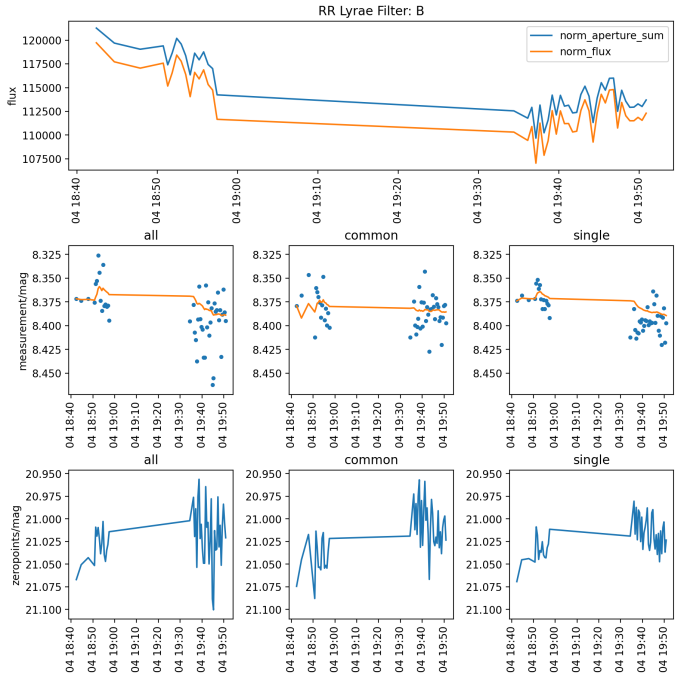

Fig. 10: Light curve of RR Lyrae in B-band. (Top): Flux normalized by exposure time without background subtraction (norm_aperture_sum) and with background subtraction (norm_flux). (Middle): Our measurements (data) with a reference light curve (model) overlayed. Individual measurements are calibrated using all reference stars (all), stars appearing in all images (common) and a single star (single). (Bottom): Calculated zero points using the different methods.

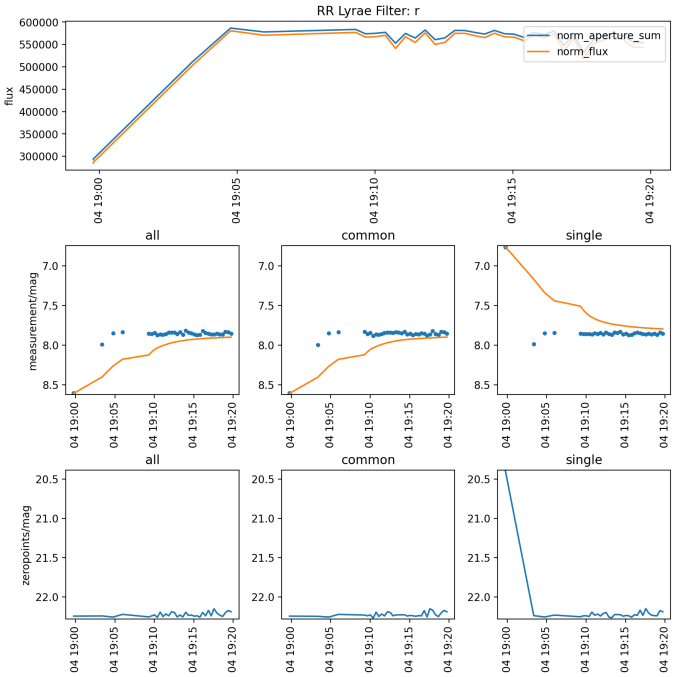

Fig. 11: Light curve of RR Lyrae in r-band. Description see Figure 10

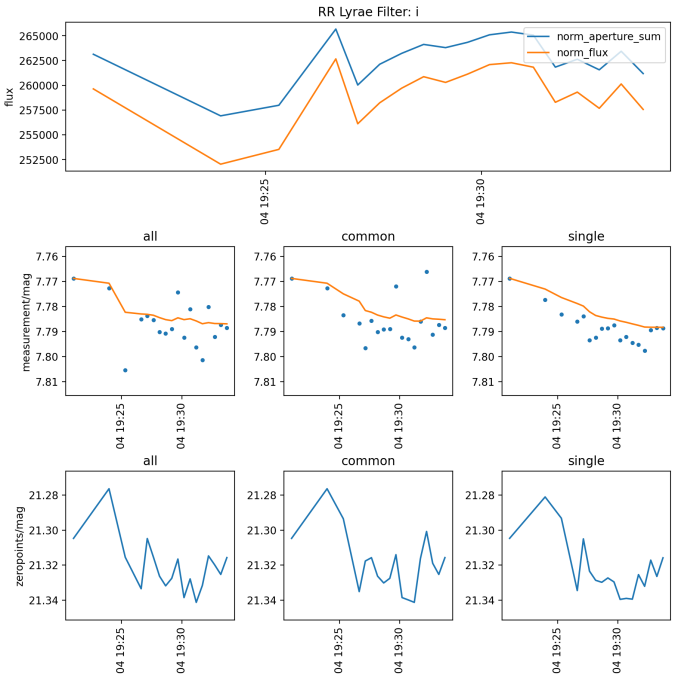

Fig. 12: Light curve of RR Lyrae in i-band. Description see Figure 10

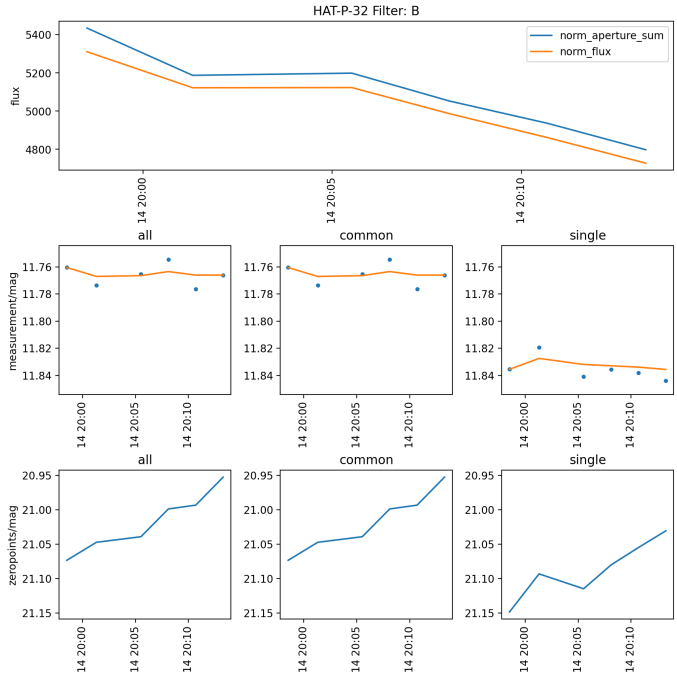

Fig. 13: Light curve of HAT-P-32 in B-band. Description see Figure 10

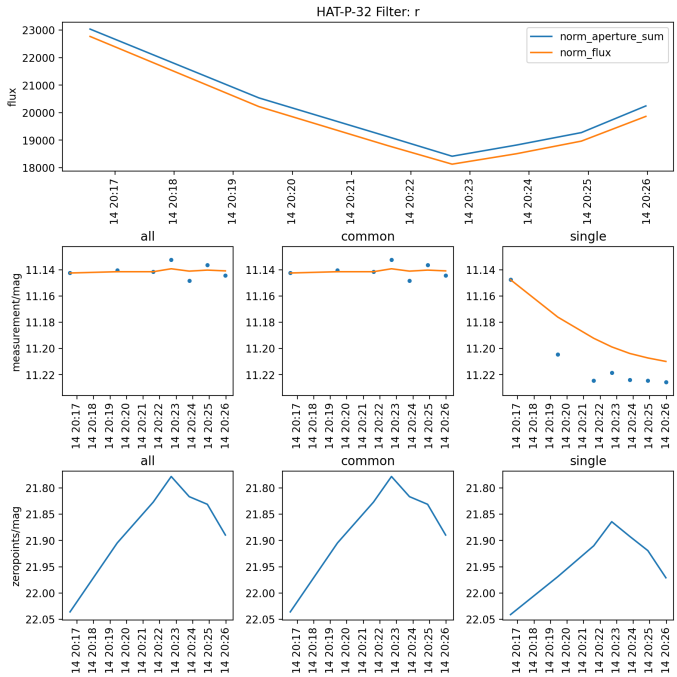

Fig. 14: Light curve of HAT-P-32 in r-band. Description see Figure 10